

Communication Protocol Effects on Multi-Agent System Efficiency and Task Completion

Final Project Presentation

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Rubric Map

FINAL PRESENTATION, 30%

Problem

Why communication protocol is an uncontrolled design choice in multi-agent LLM systems.

Contribution

A controlled benchmark, reusable code, logged artifacts, and a protocol-selection demo.

Evaluation

360 pipeline runs with token, latency, and completion metrics across three domains.

Problem

CONTEXT

Multi-agent LLM applications often pass intermediate work between specialized agents: planner, executor, integrator, critic, or retriever.

Frameworks make the communication format flexible, but rarely tell us which format is efficient or reliable.

Practical audience: developers choosing message formats in LangGraph, AutoGen, CrewAI, or custom agent pipelines.

Research Questions

SCOPE

RQ1

How much does protocol choice affect token cost and latency?

RQ2

How much does protocol choice affect task completion quality?

RQ3

Does the best protocol change across math, reading, and news?

Boundary: this is a protocol-level study, not a new task-specific agent application.

System Model

FIXED AGENT ARCHITECTURE



Held constant: model gpt-4o-mini, temperature 0.3, role prompts, domain-specific max tokens.

Manipulated: the inter-agent communication protocol only.

Protocols

EXPERIMENTAL FACTOR

NL

Plain English prose, with explicit instruction to avoid structure.

Markdown JSON

Headings and bullets for readable intermediate reasoning.

Validated JSON object with one parse-retry.

Shared Memory

A blackboard snapshot injected into every downstream agent.

Experiment

CONTROLLED GRID

4

protocols

NL, Markdown, JSON,
Shared Memory

3

domains

GSM8K, SQuAD, curated
news

360

pipeline runs

10 samples x 3 reps per cell

Efficiency: prompt tokens,
completion tokens, total tokens,
wall-clock latency.

Quality: numeric exact match,
SQuAD token-F1, and
ROUGE-based news scoring.

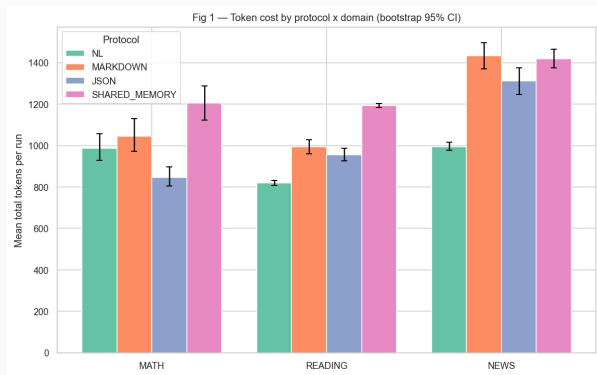
Contributions

WHAT WE BUILT

1. Reusable protocol benchmark for a fixed three-agent LLM pipeline.
2. 360-run dataset with 1,080 message-level logs.
3. ANOVA, Tukey HSD, effect sizes, and bootstrap intervals.
4. Streamlit demo for protocol recommendations.

Token Cost

RQ1: EFFICIENCY



.262

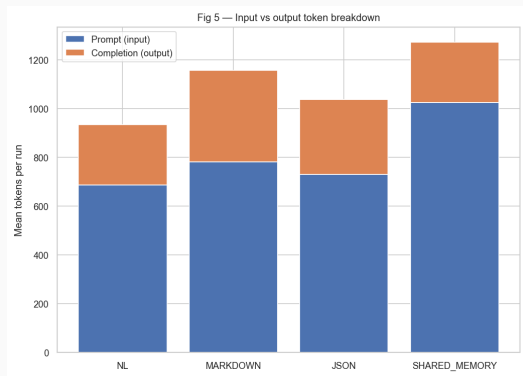
protocol effect size

two-way ANOVA on total tokens

Shared Memory is the most expensive protocol in every domain: +338 tokens/run versus NL, pooled.

Where the Tokens Go

PROMPT VS COMPLETION

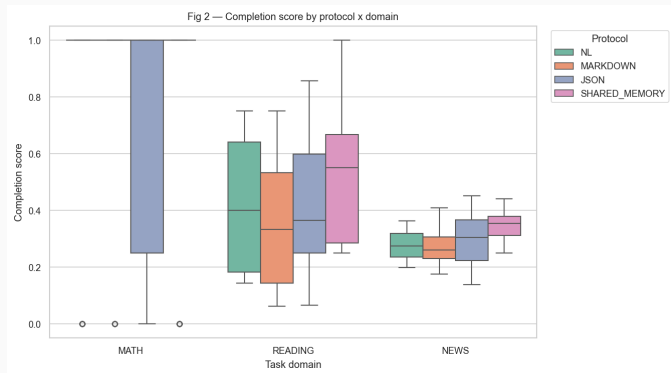


Shared Memory increases prompt tokens: every agent reads the accumulated blackboard.

Pooled cost order: NL < JSON < Markdown < Shared Memory.

Completion Quality

RQ2: EFFECTIVENESS



.017

math effect size

protocol mostly changes cost

.102

reading effect size

Shared Memory leads

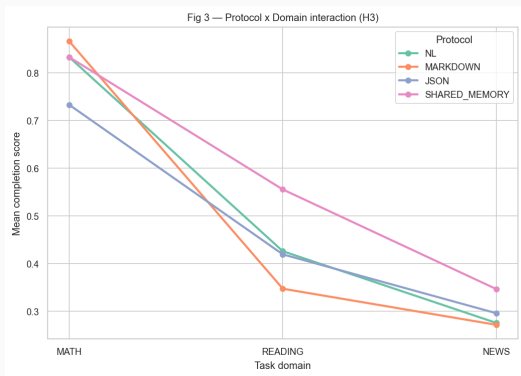
.197

news effect size

largest quality effect

Domain Interaction

RQ3: PROTOCOL BY DOMAIN

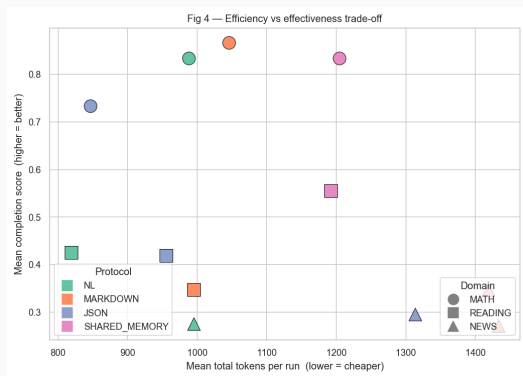


The best protocol changes across domains. Math is flat; reading and news reward richer context.

Note: pooled completion interaction: $p = .203$; token interaction: $p < .001$.

Trade-off

EFFICIENCY VS EFFECTIVENESS



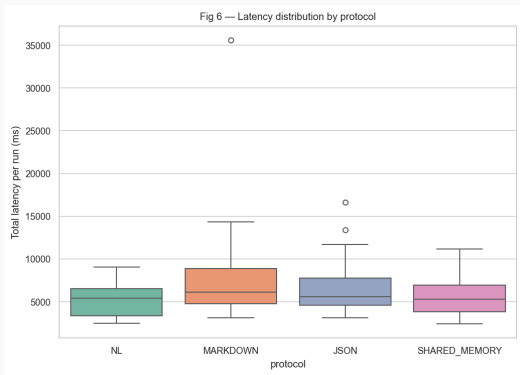
Math: Markdown for best completion; JSON for lowest cost.

Reading: Shared Memory for quality; NL for budget.

News: Shared Memory when factual completeness matters.

Latency

OPERATIONAL COST



Domain complexity dominates latency variance.

Protocol still has a measurable effect:
 $F = 16.55$, $p < .001$,
 $\eta^2 = .070$.

Token-efficient protocols tend to reduce estimated API cost.

Demo

PRACTICAL IMPACT

1. User enters a free-text task.
 2. The app classifies the task domain.
 3. It selects the empirically best protocol from
`results_summary.csv`.
 4. The pipeline reports tokens, latency, cost, and final answer.
- Artifact:** single-page Streamlit demo in `app.py`, with manual protocol override and raw message logs.

Limitations

WHAT THIS DOES NOT CLAIM

- Only 10 samples per domain and 3 repetitions per cell.
- All agents use `gpt-4o-mini`; effects may differ under other models.
- Three-agent chain only, not branching debate or tool workflows.
- Future work: larger samples, hybrid protocols, heterogeneous agents, dynamic switching.

Takeaways

FINAL RECOMMENDATION

Do not choose an agent communication protocol globally. Choose it by domain and by whether the priority is cost or completion quality.

Cost: NL and JSON are strong budget choices.

Quality: Shared Memory helps reading and news.

Artifact: code, results, figures, and demo are reproducible

Backup: Cell Summary

RESULTS SUMMARY, N=30 PER CELL

Protocol	Domain	Tokens	Prompt	Completion	Score
NL	Math	988	665	323	.833
NL	Reading	819	717	102	.426
NL	News	995	680	315	.275
Markdown	Math	1046	695	351	.867
Markdown	Reading	995	794	201	.347
Markdown	News	1433	860	574	.271
JSON	Math	847	600	247	.733
JSON	Reading	955	777	178	.419
JSON	News	1313	815	498	.296
Shared Memory	Math	1204	931	273	.833
Shared Memory	Reading	1192	1095	97	.556
Shared Memory	News	1418	1052	366	.346

Source: results/results_summary.csv